Task-1

1.Find your local IP range

```
File  Actions  Edit  View  Help

┌──(hacker㉿hacker)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.162.76  netmask 255.255.255.0  broadcast 192.168.162.255
        inet6 fe80::a00:27ff:fe88:4154  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:88:41:54  txqueuelen 1000  (Ethernet)
        RX packets 6  bytes 1750 (1.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 45  bytes 5424 (5.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(hacker㉿hacker)-[~]
└─$
```

2. To perform TCP SYN scan and open ports found.

```
┌──(hacker㉿hacker)-[~]
└─$ nmap -sS 192.168.162.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-23 19:04 IST
Nmap scan report for 192.168.162.20
Host is up (0.0014s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
MAC Address: 50:5A:65:E8:50:CD (AzureWave Technology)

Nmap scan report for 192.168.162.121
Host is up (0.0037s latency).
Not shown: 999 closed tcp ports (reset)
PORT    STATE SERVICE
53/tcp  open  domain
MAC Address: B6:64:4C:4E:7F:D3 (Unknown)

Nmap scan report for 192.168.162.76
Host is up (0.000015s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 10.32 seconds

┌──(hacker㉿hacker)-[~]
└─$
```

3 .Research common services running on those ports.

```
┌──(hacker㉿hacker)-[~]
└─$ sudo nmap -sS 192.168.162.1-100
[sudo] password for hacker:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-23 19:13 IST
Nmap scan report for 192.168.162.20
Host is up (0.0013s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
MAC Address: 50:5A:65:E8:50:CD (AzureWave Technology)

Nmap scan report for 192.168.162.76
Host is up (0.000012s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh

Nmap done: 100 IP addresses (2 hosts up) scanned in 9.15 seconds
┌──(hacker㉿hacker)-[~]
└─$
```

4. Identify potential security risks from open ports

Port	State	Service	⚠ Potential Risk
135	open	MSRPC	Might be leveraged for remote exploits (e.g., RPC DCOM attacks), reconnaissance, or worm propagation if unpatched.
139	open	NetBIOS-SSN	Enables legacy file/printer sharing over NetBIOS. Susceptible to brute force attacks, information disclosure, and exploits (e.g., EternalBlue).
445	open	Microsoft-DS	Main SMB port. Frequently targeted by ransomware and worms (e.g., MS17-010/ EternalBlue). Must be patched and access restricted.
3306	open	MySQL	Default database port. Risk of brute force attacks, SQL injection, and unauthorized data access if not secured properly.
MAC	AzureWave Tech	-	Suggests it's a device (perhaps an IoT device or server) that may have weaker security measures.

Port	State	Service	⚠ Potential Risk
21	open	FTP	Insecure if using plain FTP (no encryption). Vulnerable to brute force attacks, data sniffing, and anonymous access exploits.
22	open	SSH	Generally secure when configured properly. Risk if using weak passwords or outdated SSH versions.